Earth System

Earth System Science

Chapter 1: Introduction

Objective

- *To Define earth's system*
- *To Know The earth's system components*
- *To be aware Energy and Matter in the Earth System*

INTRODUCTION

What is a system?

- A system is a set of components that work together or interact.
- A bicycle is a good example of a system. *The frame, wheels, pedals, brakes, and other components work together to make the bicycle a vehicle.*
- Earth, from the very top of the atmosphere to the core at its center, is considered as a system.

The Earth System comprises of four major components:

- The Atmosphere,
- The Hydrosphere,
- The Geosphere and
- The Biosphere.

These spheres work together in very complex ways to make the Earth the place we know it today.

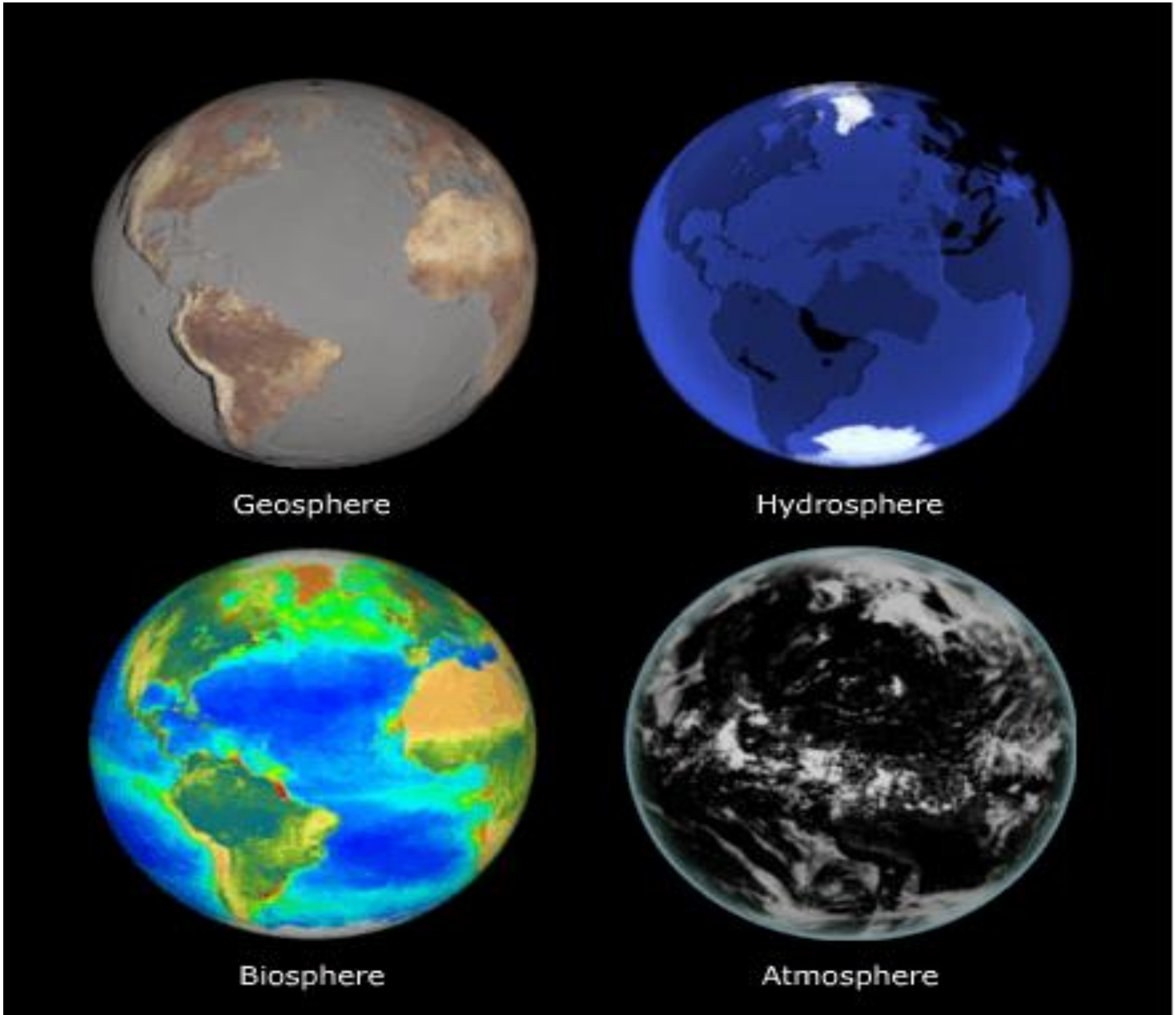
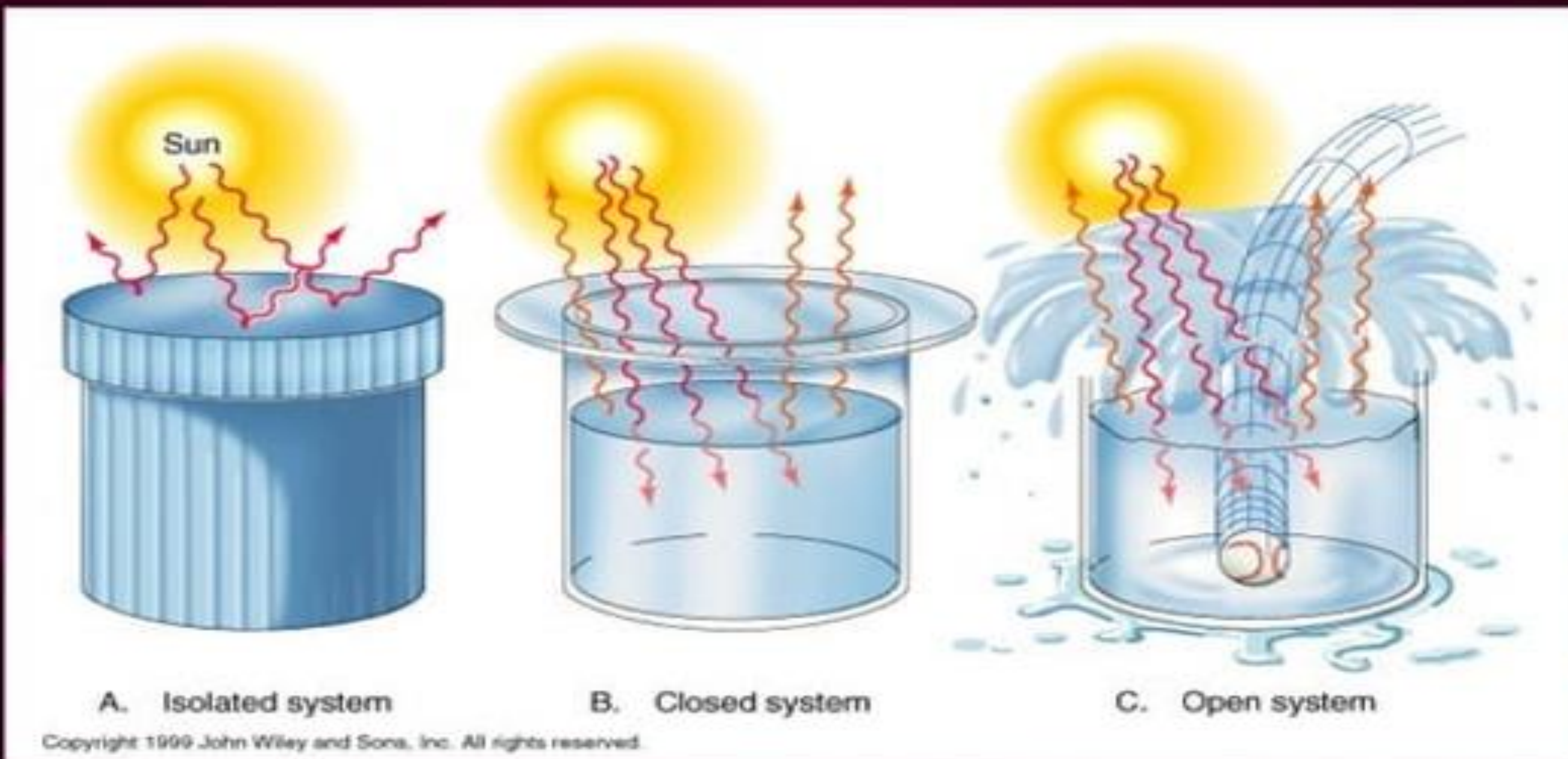


Image credits: Jennifer Loomis, TERC/Biosphere image provided by ORBIMAGE © Orbital Imaging Corporation. Processing by NASA Goddard Space Flight Center. Image source: Exploring Earth

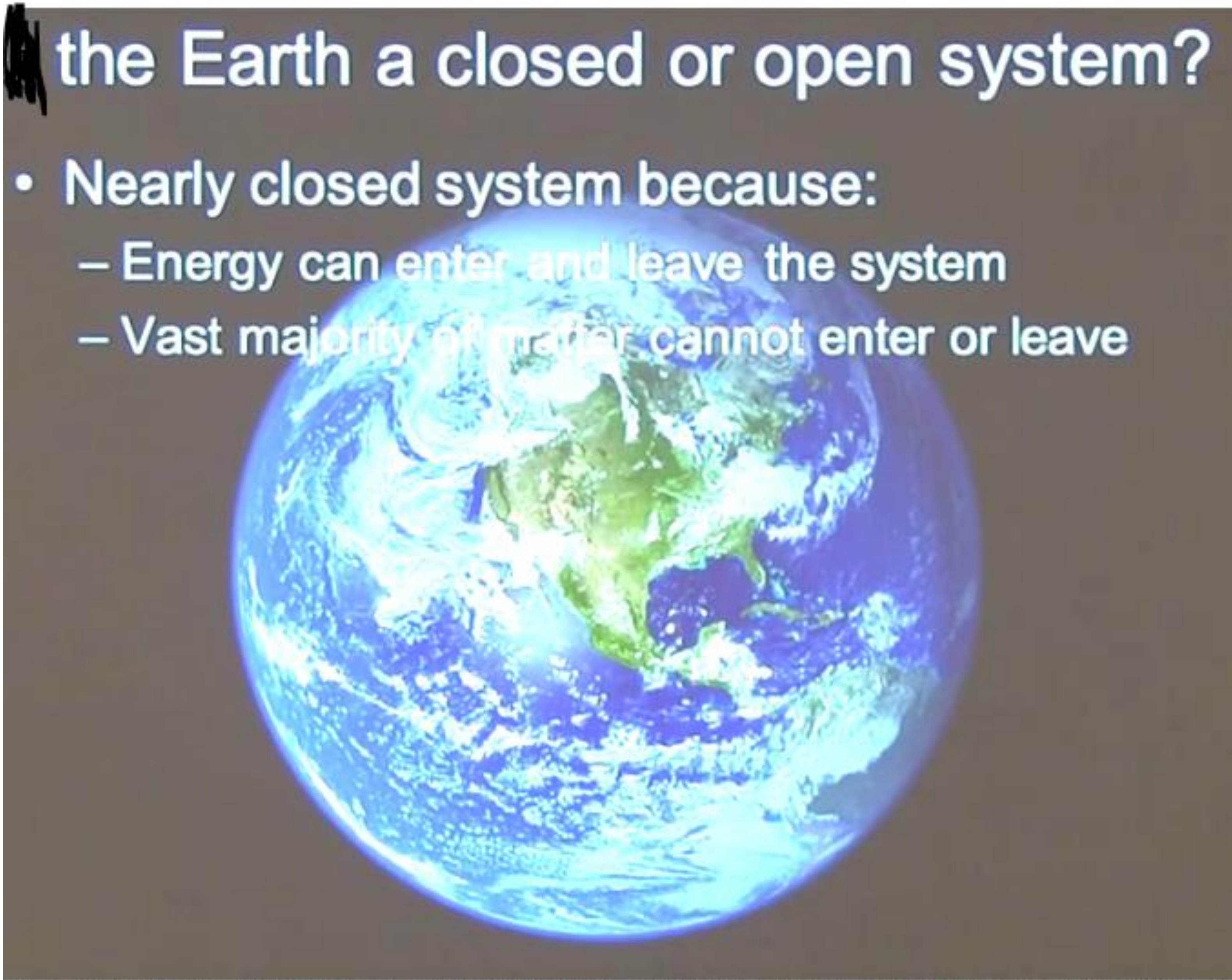
Systems: Open vs Closed

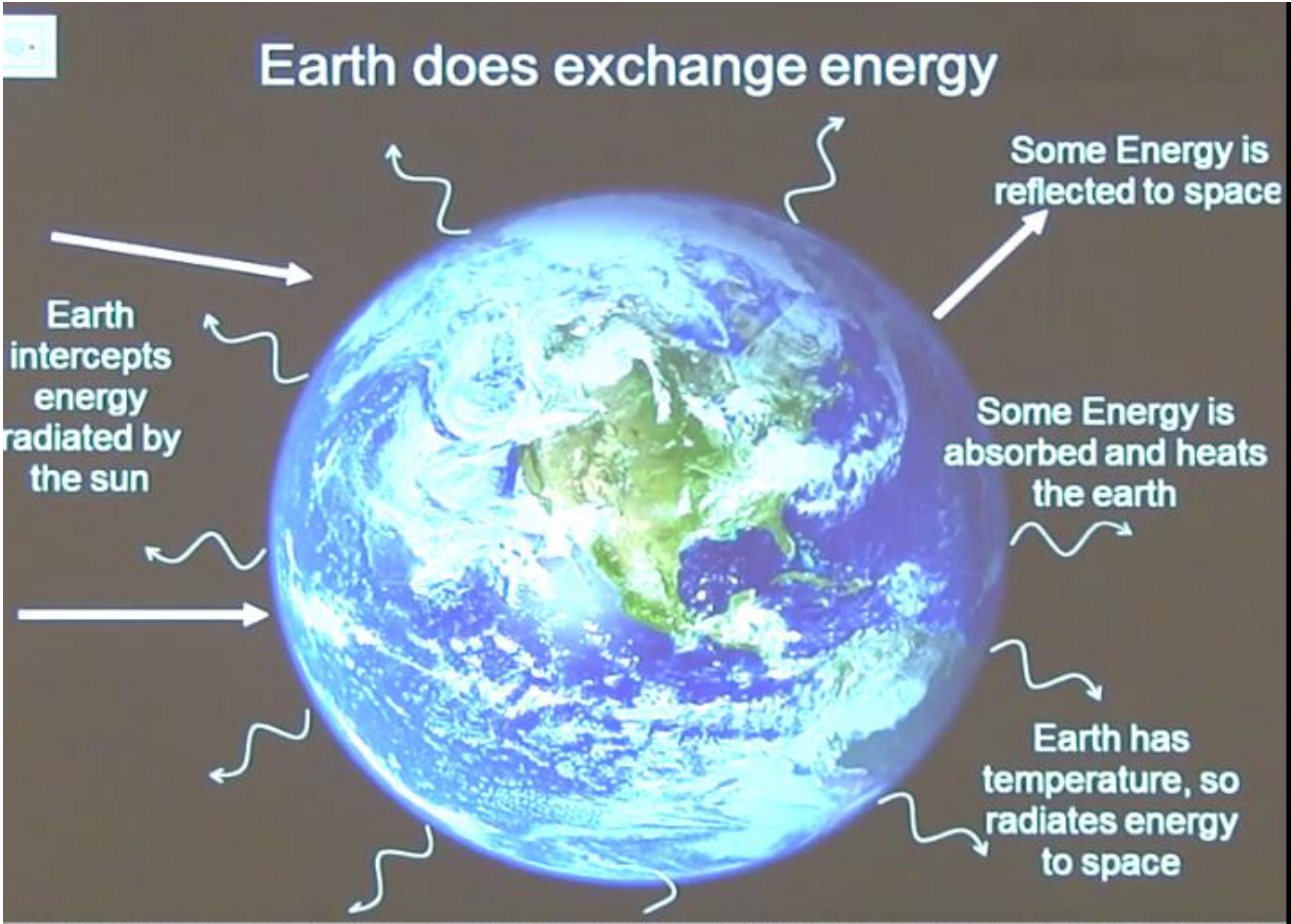


Closed system: exchange of energy but negligible exchange of mass with surroundings

[redacted] the Earth a closed or open system?

a) Open
b) Closed
c) Neither


[redacted] the Earth a closed or open system?
• Nearly closed system because:
– Energy can enter and leave the system
– Vast majority of matter cannot enter or leave


Earth does exchange energy
Some Energy is reflected to space
Earth intercepts energy radiated by the sun
Some Energy is absorbed and heats the earth
Earth has temperature, so radiates energy to space


Earth only exchanges tiny amount of matter
In:
Comets
Meteorites
Out:
Hydrogen
Helium

Closed System – Implications?

- Resources are finite
- Can't throw things 'away'
- Changes in one part of a closed system will eventually affect the other parts

Why Earth Science?

- Because nearly everything we do each day is connected in some way to the Earth: to its land, oceans, atmosphere, plants, and animals

- Because to predict the weather as accurate as possible we need to have a solid understanding about the interconnection between earth system components.

- Because we LIVE on Earth!!

What is Earth System?

- The term Earth System refers to the suite of interacting physical, chemical and biological global-scale cycles and energy fluxes that provide the **life-support system for life** at the surface of the planet.

- *This definition of the Earth System goes well beyond the notion that the geophysical processes encompassing the Earth's great fluids—the ocean and the atmosphere—generate the planetary life-support system on their own.*

- In our definition, biological / ecological processes are an integral part of the functioning of the Earth System and not merely the recipient of changes in the coupled ocean-atmosphere part of the system.

- A second critical feature is that **forcings and feedbacks within the Earth System are as important as external drivers of change**, such as the flux of energy from the sun.

- Finally, the Earth System **includes humans, our societies, and our activities**

- *Thus humans are not an outside force perturbing an otherwise natural system but rather an integral and interacting part of the Earth System itself.*


Atmosphere
Anthroposphere
Policy
Economy
Society

- The outer boundary of the Earth System is the top of the atmosphere.
- The sun is outside the Earth system.
  - ✓ *It provides our main source of energy but it is not affected by what goes on within the components of the earth system.*

Earth System Science (ESS)

- The study of the interactions between and among events and Earth’s spheres

The Rise of Earth System Science

- Advances in technology allow scientists to see Earth in new ways.

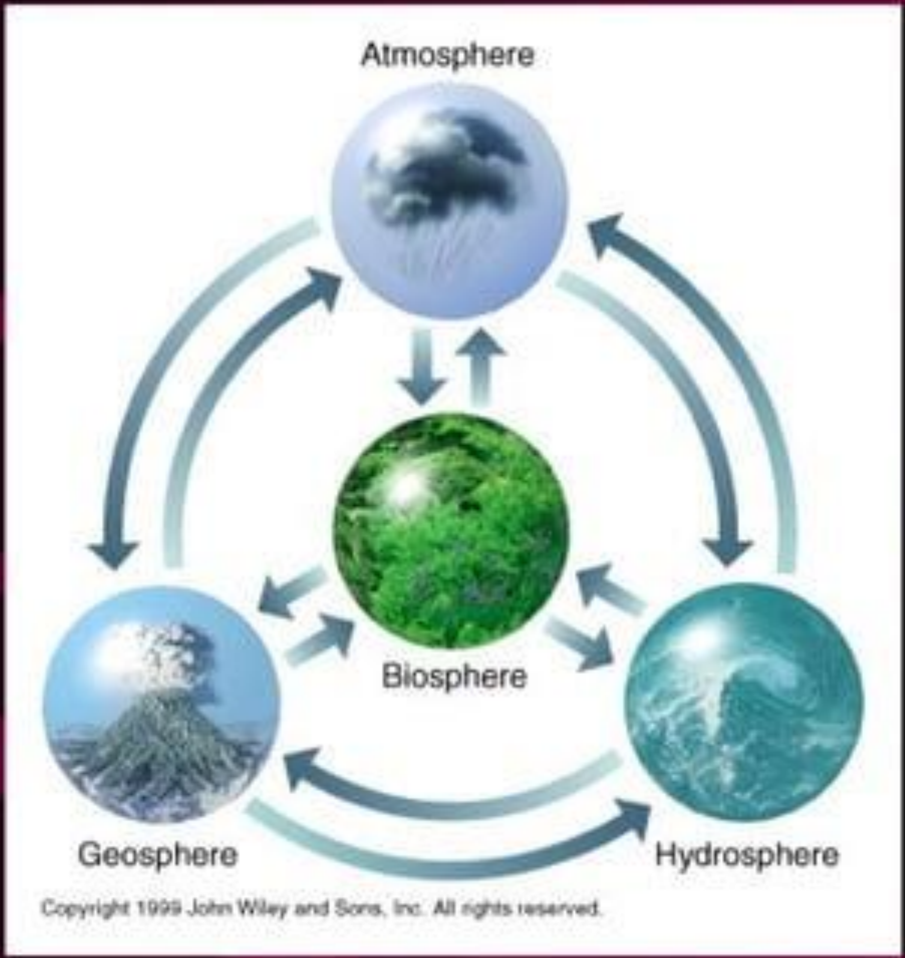
- Satellites let us see Earth from space.

- We also use submersibles to explore ocean depths.

What is the source of energy that drive Earth’s systems?

The earth's system components

The Earth's Four Spheres



The Atmosphere

this is the gaseous layer surrounding the earth and held to its surface by gravity.

Elements of the atmosphere include several different gases (nitrogen and oxygen in the greatest abundance) as well as clouds, water vapor, and solid particles such as dust and pollen.

The Hydrosphere

this consists of those parts of the earth system composed of water in its liquid, gaseous (vapor) and solid (ice) phases.

Elements of the hydrosphere include all bodies of surface water as well as ground water.

Earth's solid water is part of the hydrosphere but is also considered a sub-sphere with its own name: the *Cryosphere*.

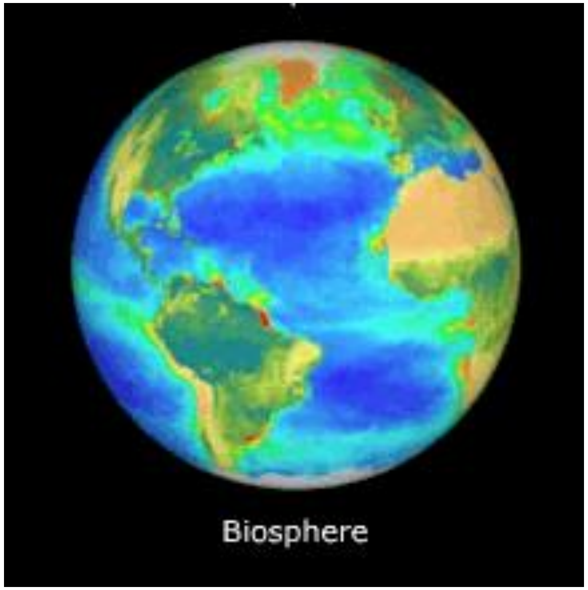
The Geosphere

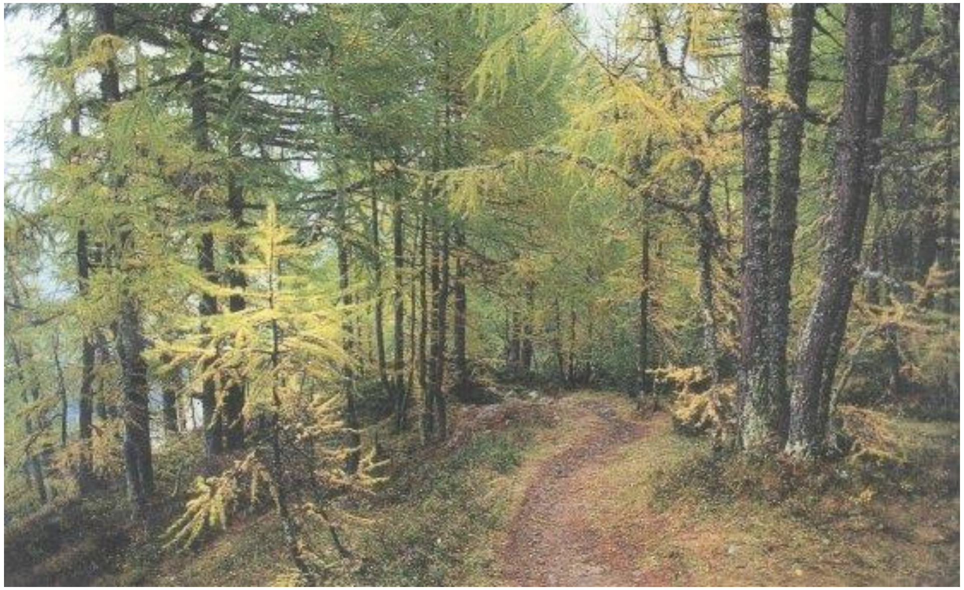
this is the part of the planet composed of rock and minerals; it includes the solid crust, the molten mantle and the liquid and solid parts of the earth’s core.

Elements of the geosphere include soils.

Soil is a sub-sphere of the geosphere with its own name: the *pedosphere*.

The Biosphere





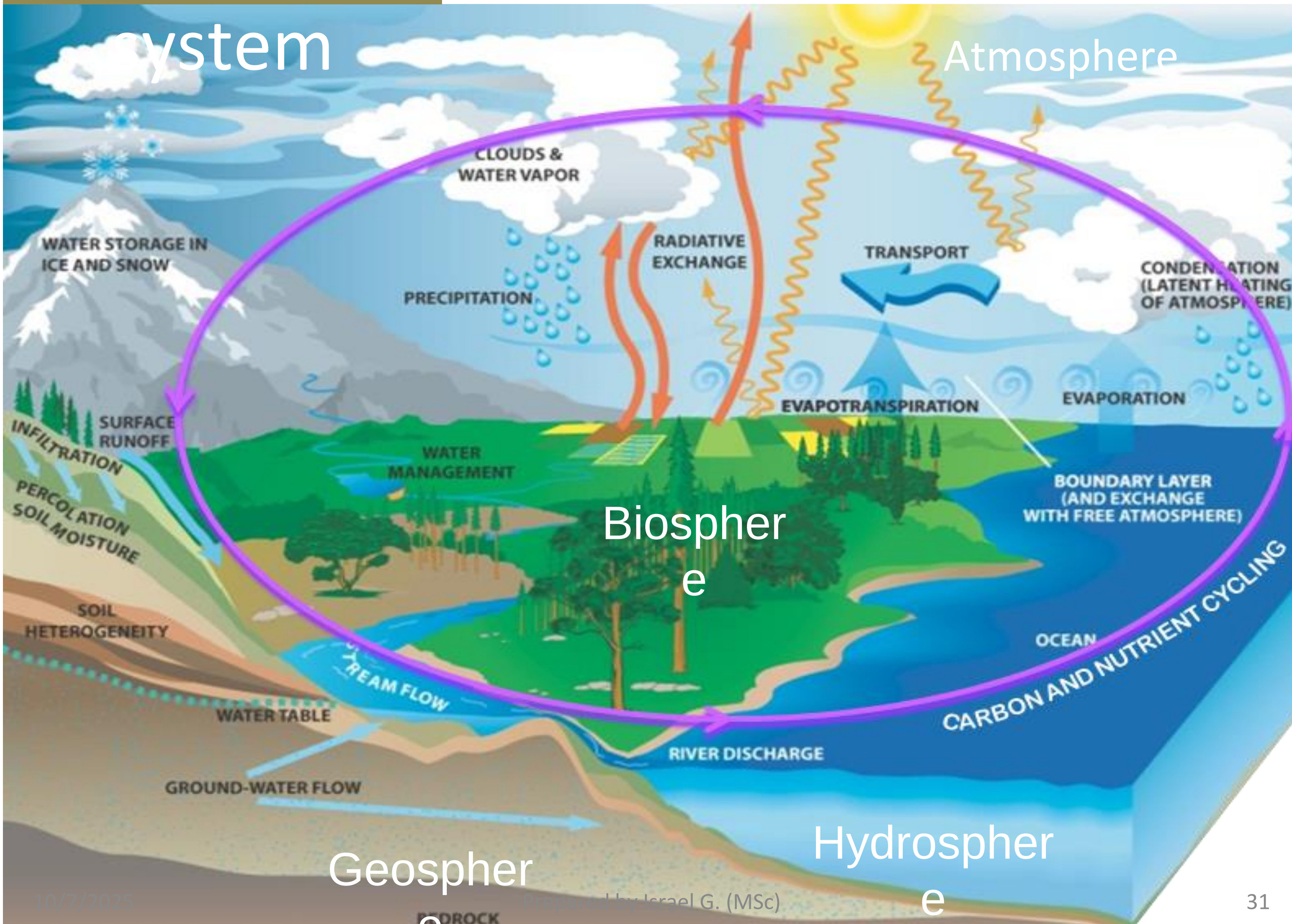
Elements of the biosphere include all living plants and animals, from the smallest single celled organisms to the largest trees and animals on the planet.

Which of the four major spheres do you see in this image? Explain your answers.

Which of the four major spheres do you see in this image? Explain your answers.

Wikimedia commons: public domain

Earth system



A key concept of the Earth system is that matter and energy cycle constantly between spheres and between elements of a sphere.

What interconnections between the spheres do you think are happening here?

A key concept of the Earth system is that matter and energy cycle constantly between spheres and between elements of a sphere.

What is happening here?

A key concept of the Earth system is that matter and energy cycle constantly between spheres and between elements of a sphere.

What is happening here?

Energy and Matter in the Earth System

- *Energy:* The Sun is the primary source, driving weather, climate, and life processes.
- *Matter:* Elements such as carbon, nitrogen, and water cycle between the different subsystems.

The Earth System operates through the continuous flow of energy and the cycling of matter, maintaining conditions suitable for life

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Thanks !

Earth System Science

Chapter 1: Introduction

Objective

- *To know the examples of earth system processes*
- *To be aware of feedback loops*
- *To identify examples of feedback loops*

Examples of Earth System Processes

- Evaporation and precipitation (linking hydrosphere and atmosphere).
- Photosynthesis (linking biosphere and atmosphere)
- Volcanic outgassing (linking geosphere and atmosphere).
- Earth functions as an interconnected system where change in one part affects all others

The Feedback Mechanisms in Earth System Science

Feedback Definition

- Feedback refers to a chain of cause and effect that forms a closed group.
- It is a process in which a change in one part of the Earth system causes effects that either amplify or reduce that change.
- It control the stability and sensitivity of Earth's climate and weather systems.

Types of Feedback

There are two types of feedback: *Positive* and *Negative*.

Positive feedback is an amplifying loop of causal connections.

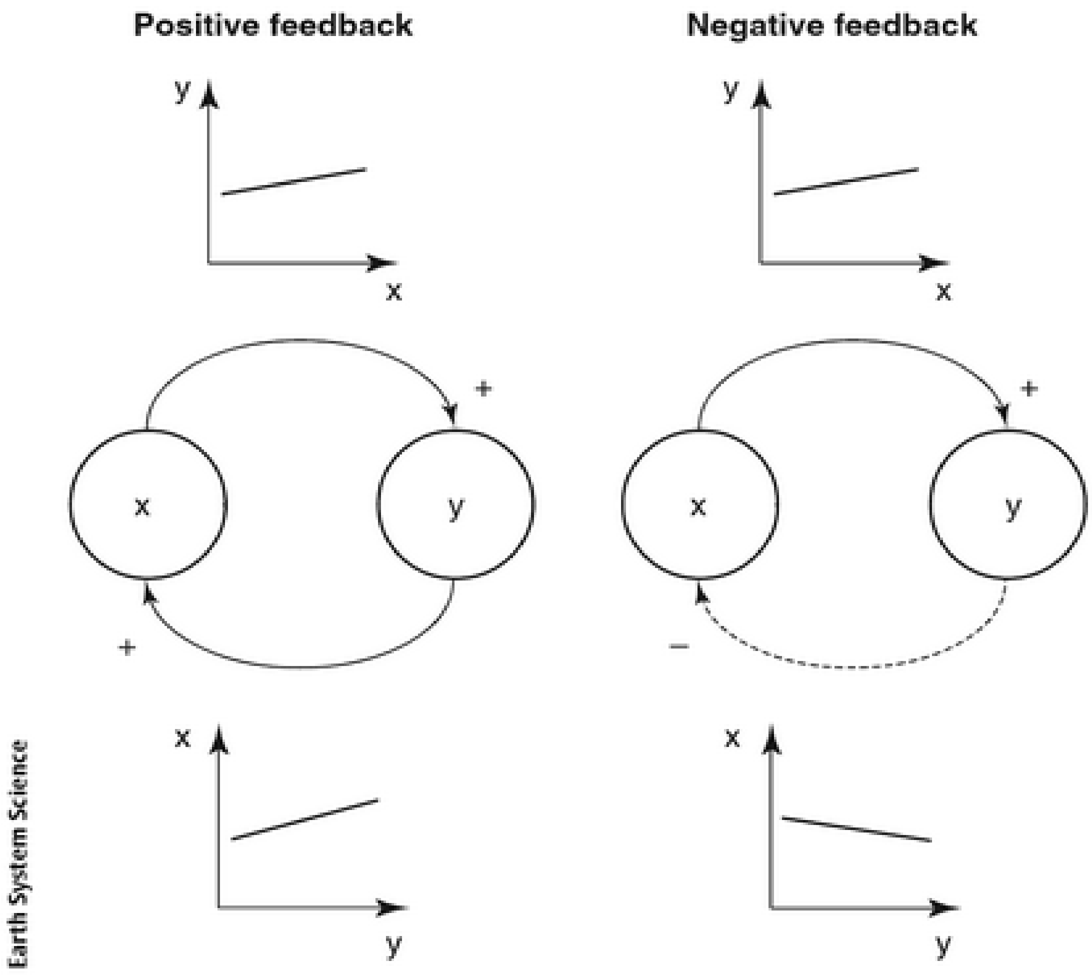
> *This means an initial perturbation to any part of the loop will trigger a response that amplifies the initial change.*

Types of Feedback

Negative feedback is a damping loop of causal connections.

This means an initial perturbation to any part of the loop will trigger a response that damps the initial change.

Types of Feedback



2. Positive and negative feedback. A plus sign on a solid arrow indicates a direct relationship (e.g. increasing x increases y). A minus sign on a dashed arrow indicates an inverse relationship (e.g. increasing y decreases x). An even number (including zero) of inverse relationships in a loop gives a positive feedback, and an odd number gives a negative feedback.

Positive Feedback

- Process that reinforces an initial change, making it stronger.

Impact in Meteorology

- Enhances warming trends, intensifies extreme weather events.

Negative Feedback

- Process that opposes an initial change, reducing its effect.

Impact in Meteorology

- Stabilizes climate, regulates temperature, and moderates extreme weather

- Identifying the feedback loops in the earth system and understanding the behavior they can create is a key part of earth system science.

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Thanks !